

Project Statement: Student Performance Analyzer

Problem Context:

A school wants to analyze the academic performance of its students in different subjects. The school management has provided a dataset containing student details (name, roll number, age, class, subjects, and marks). You need to build a Python project that can **analyze, process, and generate insights** from this data using Python, NumPy, Pandas, and OOP concepts. You also need to make the project robust with proper **exception handling**.

Project Requirements:

1. Data Handling with Python Basics

- Accept student details (like name, roll no, age, class, marks) from the user.
- Store the data in a structured way (list/dictionary).
- Display the entered details in a clean format.
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2. Use of NumPy (Basic)

- Use NumPy arrays to store and calculate marks.
- Perform basic operations like:
 - Calculate average marks.
 - Find the highest and lowest marks in a subject.
 - Calculate overall class performance (mean, median, standard deviation).

3. Use of Pandas

- Load the student data into a Pandas **DataFrame**.
- Perform operations like:
 - Display top 5 and bottom 5 students based on marks.
 - Group students by class and calculate average marks.
 - Save the final report into a CSV/Excel file.

4. OOP Concepts

- Create a Student class with attributes like name, roll_no, age, marks.
- Create methods to:
 - Display student details.
 - Calculate grade (A, B, C, etc.) based on average marks.
- Create a Classroom class to store multiple students and perform operations like adding/removing students, calculating overall performance.

5. Exception Handling

- Handle errors such as:
 - Invalid input (e.g., entering string instead of number).
 - Division by zero (in case of wrong calculations).
 - File not found errors when loading/saving CSV.
- A Python program that:
 1. Accepts and stores student data.
 2. Performs analysis using NumPy & Pandas.
 3. Implements OOP for better structure.
 4. Handles exceptions gracefully.